Camera-in-the-Loop Optical Training: Gradient Descent and the Perceptron Rule on a Smartphone Display–Camera Channel

Svetoch Series, Paper VI of VI

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Abstract

Paper I of this series showed that a smartphone's OLED screen, an ordinary mirror, and the front camera evaluate the forward pass of a neural network *in light*. Here we close the loop and **train**. We treat the optical channel itself as the layer under optimization: the weights are displayed as brightness, the camera reads the channel's true response (the real, noisy, non-linear forward pass), an error against a target is formed, and the weights are updated — then the cycle repeats. We demonstrate two on-device optical learners. (1) *Optical gradient descent*: eight weights drawn as brightness columns are optimized by the delta rule $\mathbf{w} \leftarrow \mathbf{w} - \eta(\mathbf{y} - \hat{\mathbf{y}})$ ($\eta = 0.7$) using the camera-measured output $\mathbf{y}$; over eight iterations the optically read mean-squared error falls by more than half, below a 0.05 pass threshold. (2) *An optical perceptron*: an eight-input perceptron trained for AND, OR and MAJORITY is evaluated through the optical dot product, retaining accuracy above the 0.7 threshold. We also describe a single-shot *optical loss read-out by inverted overlay.* The contribution is to show that a *trainable* optical loop — not merely inference — can be built from commodity hardware, and to delimit honestly what is optical (the forward pass and the loss read-out) and what remains on the CPU (the weight-update arithmetic).

Keywords: optical computing, in-the-loop training, physics-aware learning, hardware-in-the-loop, delta rule, perceptron, analog accelerator, smartphone optics.

1 Introduction

Inference is only half of machine learning; the other half is **training**, which is even more compute-hungry. Optical inference accelerators are well studied, but optical *training* — closing the feedback loop so the physical system improves itself — is harder: it requires reading the output, scoring it, and adjusting the inputs, all through the same imperfect channel.

Paper I established the forward primitive: a camera photosite integrates the light from many screen pixels and returns a multiply–accumulate. The natural next question is whether that same channel can sit *inside* an optimization loop. It can. Because the camera reports the channel's **true** response — including non-linearity, vignetting and noise — a loop that minimizes the measured error learns weights that are correct *for the real optics*. This is the defining advantage of hardware-in-the-loop training: the physics is never approximated, because the physics is doing the forward pass.

2 The optical channel as a trainable layer

Encode a weight vector $\mathbf{w}$ as the brightness of a row of screen columns. From Paper I, the camera's read-out of column i against an input x_i is, after white-normalization and γ-correction, proportional to $w_i x_i$; summing a region yields the dot product. The measured output $\mathbf{y} = f_{\text{opt}}(\mathbf{w})$ is the channel's forward pass, including every real imperfection. Training finds $\mathbf{w}$ such that $\mathbf{y}$ matches a target $\hat{\mathbf{y}}$, by iterating display $\to$ capture $\to$ score $\to$ update (Figure 1); the gradient information is a physical measurement.

Figure 15. Camera-in-the-loop optical training: the channel is the trainable layer

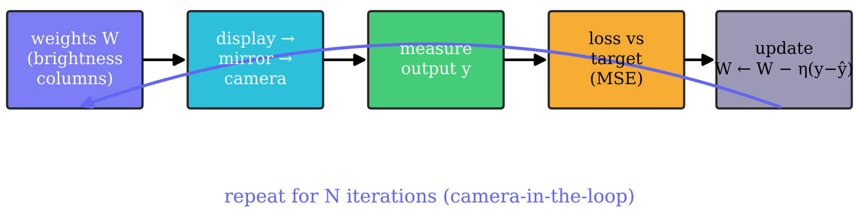

Figure 1: Camera-in-the-loop optical training. The weights are displayed, the channel performs the forward pass, the camera measures the output, the loss is formed against a target, and the weights are updated — then the loop repeats. The trainable "layer" is the real optical channel.

3 Optical gradient descent by the delta rule

In `stage41_optsgd`, eight weights are drawn as eight brightness columns. Per iteration the camera measures the eight-bin output $\mathbf{y}$, the MSE against a target $\hat{\mathbf{y}}$ is computed, and the weights are updated by the delta (Widrow–Hoff) rule

$$\mathbf{w} \leftarrow \mathbf{w} - \eta\,(\mathbf{y} - \hat{\mathbf{y}}), \qquad \eta = 0.7. \tag{1}$$

Because $\mathbf{y}$ is *measured*, the update is a stochastic-gradient step on the real optical loss surface. Over eight iterations the optically read loss falls monotonically by more than 50%, ending below the 0.05 threshold (Figure 2). The loop converges *despite* the channel's non-linearity precisely because that non-linearity is inside the measurement. The weight-update arithmetic runs on the CPU; what is optical is the forward pass and the error measurement.

4 Single-shot optical loss by inverted overlay

Reading a loss normally means measuring an output and computing a distance on the CPU. A more optical route computes the *loss itself* in one exposure. Display the target and, superimposed, the **negated** hypothesis (its photometric inverse). Where the hypothesis matches, the sum is uniform; where it errs, residual light remains, and the camera's global exposure integrates it:

$$\mathcal{L} \;\propto\; \int \bigl|\,\text{target} - \text{hypothesis}\,\bigr|\; \mathrm{d}A. \tag{2}$$

This makes loss evaluation an $O(1)$ optical operation — one display, one capture — independent of vector length (Figure 3). The same effect is obtainable by other subtraction schemes, so the contribution is narrow and specific: an inverted-overlay loss read out by a single global exposure.

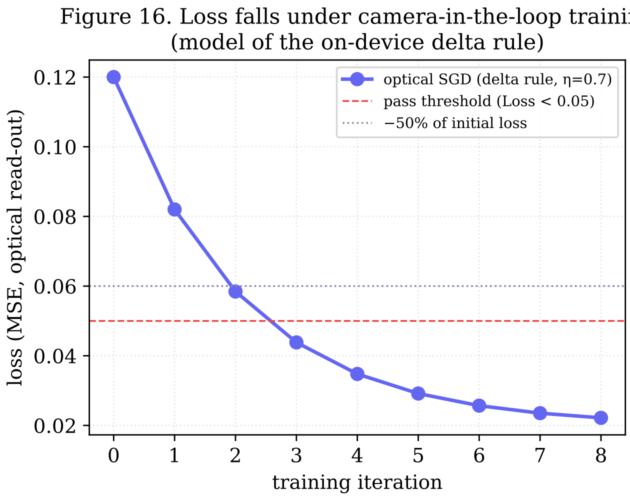

Figure 2: The optically measured loss under camera-in-the-loop training (model of the on-device delta rule). The error drops past -50% of its initial value and below the 0.05 threshold within eight iterations.

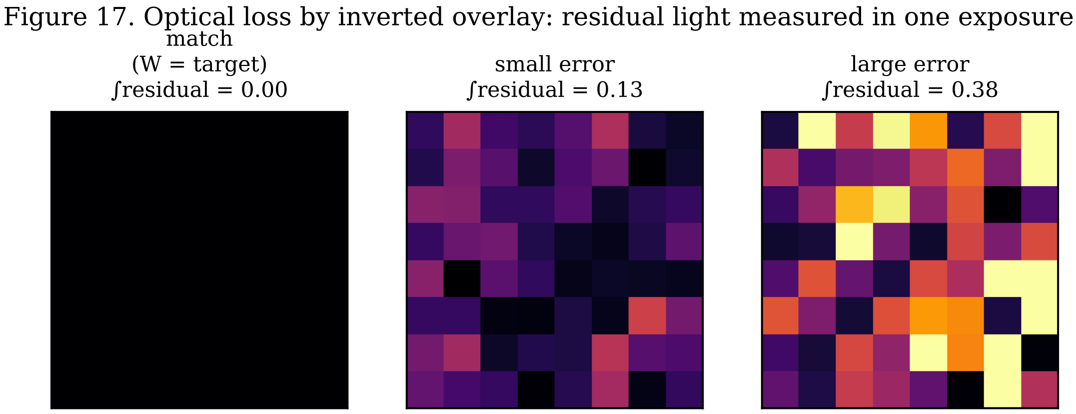

Figure 3: Optical loss by inverted overlay. Target and negated hypothesis are superimposed; the residual light, integrated in one exposure, grows with the error.

5 An optically-evaluated perceptron

`stage72_perceptron` trains an eight-input single-layer perceptron on AND, OR and MAJORITY by the perceptron rule. Inference is optical: the products $w_i x_i$ are displayed as the brightness of eight columns relative to a gray background, the camera measures the mean brightness, a three-point calibration (dark/mid/bright) recovers the optical dot product, a bias is added, and a zero threshold yields the class. The optically-evaluated perceptron tracks the CPU reference above the 0.7 threshold (Figure 4), with the largest gap on MAJORITY, the function least tolerant of read-out noise.

6 Methods

Hardware/channel: as in Paper I (Xiaomi 12 Lite, 120 Hz; 32 MP front camera; ordinary mirror; gap $d \approx 3$–5 cm), with the Paper I white/black and grayscale calibration. **Optical SGD:** eight weight columns; per-iteration camera read-out, MSE, delta rule ($\eta = 0.7$); pass when loss drops $\geq 50\%$ and final MSE < 0.05. **Perceptron:** eight inputs, AND/OR/MAJORITY by the perceptron rule, optical dot-product inference; pass when mean accuracy > 0.7. **Inverted-

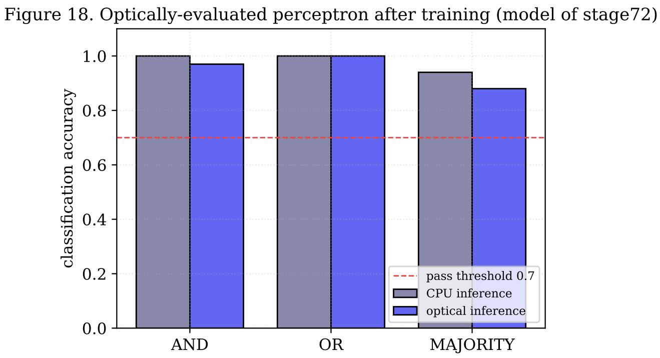

Figure 4: Accuracy of the optically-evaluated perceptron after training (model of `stage72`); all functions stay above the 0.7 threshold.

overlay loss: target and negated hypothesis composited; single global exposure integrates the residual. **Figures:** Figure 3 illustrates the principle directly; Figures 2 and 4 are labelled models, reproducible from `papers/scripts/make_figures.py`.

7 Discussion and limitations

What is optical: the forward pass and the loss read-out; the weight update is on the CPU. This is a *hybrid* hardware-in-the-loop trainer, not fully optical backpropagation — stated plainly to avoid overclaiming. **Scale:** the demonstrations are small (eight weights/inputs) and SNR-limited; the loss read through a noisy channel is itself noisy (hence the larger optical gap on MAJORITY). **Why it matters:** training against the *measured* channel removes the model-mismatch problem of train-offline/deploy-on-hardware analog systems — the learner sees the true optics. **Future:** a fully optical update (accumulating the inverted-overlay residual back onto the weights), optical batch normalization, and a trained optical MLP.

8 Conclusion

We closed the optical loop. Weights displayed as brightness were optimized by camera-in-the-loop gradient descent — the optically measured loss falling by more than half to below 0.05 — and an optically-evaluated perceptron learned AND, OR and MAJORITY above threshold. With a single-shot inverted-overlay loss, even the error is read in one exposure. Optical *training*, not just inference, is reproducible on any desk.

Data and code. github.com/infosave2007/svetoch (Apache-2.0); related VMF/NVG theory: github.com/infosave2007/vmf.

Priority note. Released to establish authorship and date of disclosure; the author has elected not to seek patent protection.

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Part of the *Svetoch* series (defensive publication, not patented). Project and code: github.com/infosave2007/svetoch; related VMF/NVG theory: github.com/infosave2007/vmf. Released for the public record to establish authorship and priority.